# Different types of Neural Networks in Deep Learning

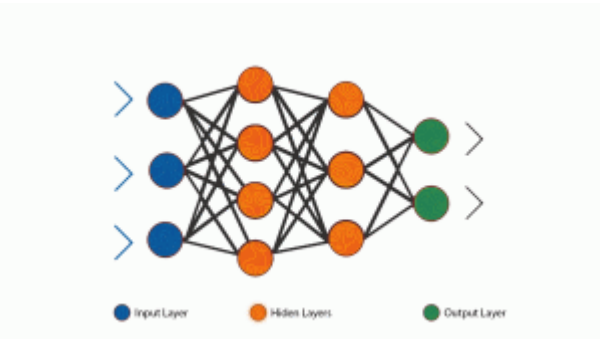
This article focuses on three important types of neural networks that form the basis for most pre-trained models in deep learning:

- Artificial Neural Networks (ANN)
- Convolution Neural Networks (CNN)
- Recurrent Neural Networks (RNN)

Let's discuss each neural network in detail.

## Artificial Neural Network (ANN) – What is a ANN and why should you use it?

A single perceptron (or neuron) can be imagined as a Logistic Regression. Artificial Neural Network, or ANN, is a group of multiple perceptrons/ neurons at each layer. ANN is also known as a **Feed-Forward Neural network** because inputs are processed only in the forward direction:




*ANN*

As you can see here, ANN consists of 3 layers – Input, Hidden and Output. The input layer accepts the inputs, the hidden layer processes the inputs, and the output layer produces the result. Essentially, each layer tries to learn certain weights.

*If you want to explore more about how ANN works, I recommend going through the below article:*

- *Understanding and Coding Neural Networks From Scratch in Python and R*

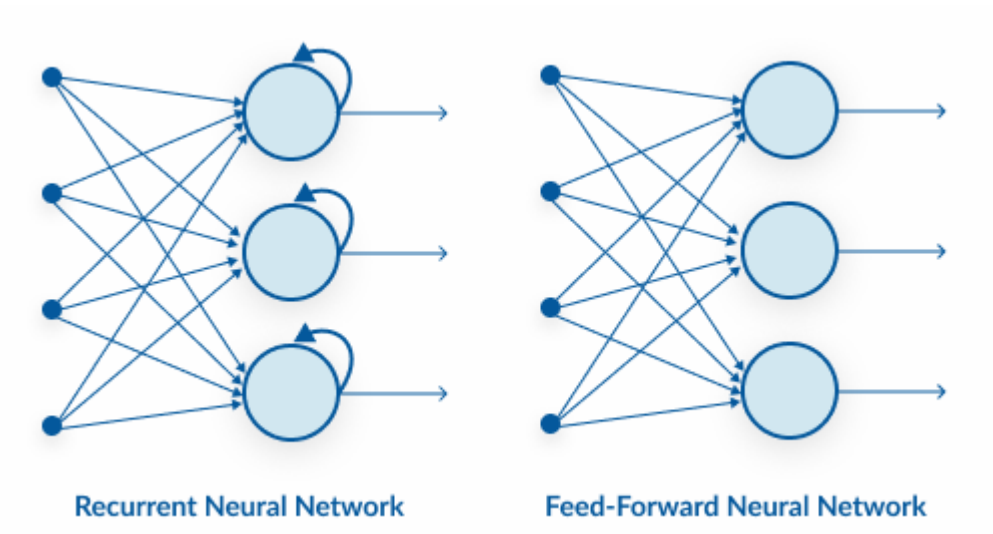
ANN can be used to solve problems related to:

- Tabular data
- Image data
- Text data

## Recurrent Neural Network (RNN) – What is an RNN and why should you use it?

Let us first try to understand the difference between an RNN and an ANN from the architecture perspective:

A looping constraint on the hidden layer of ANN turns to RNN.




**As you can see here, RNN has a recurrent connection on the hidden state. This looping constraint ensures that sequential information is captured in the input data.**

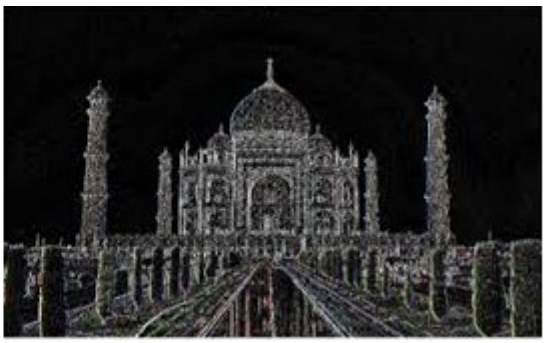
We can use recurrent neural networks to solve the problems related to:

- Time Series data
- Text data
- Audio data

## Convolution Neural Network (CNN) – What is a CNN and Why Should we use it?

Convolutional neural networks (CNN) are all the rage in the deep learning community right now. These CNN models are being used across different applications and domains, and they're especially prevalent in image and video processing projects.

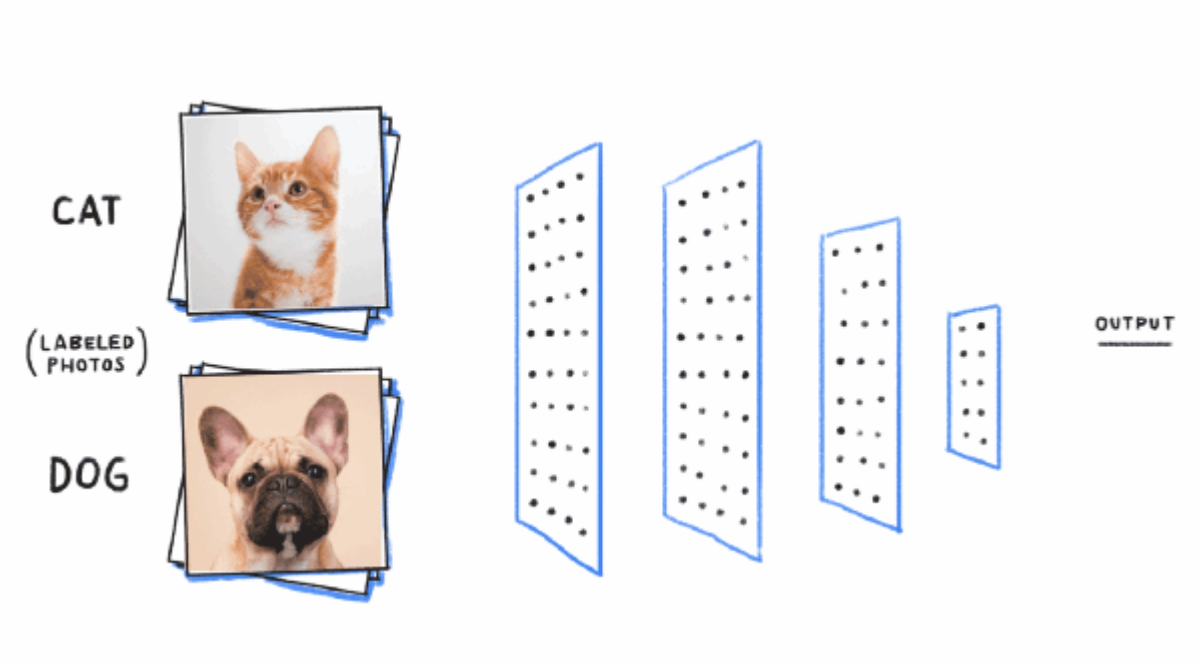
**The building blocks of CNNs are filters a.k.a. kernels.** Kernels are used to extract the relevant features from the input using the convolution operation. Let's try to grasp the importance of filters using images as input data. Convolving an image with filters results in a feature map:



*Output of Convolution*

Advantages of Convolution Neural Network (CNN)

- CNN learns the filters automatically without mentioning it explicitly. These filters help in extracting the right and relevant features from the input data




*CNN – Image Classification*

Comparing the Different Types of Neural Networks (MLP(ANN) vs. RNN vs. CNN)

| | MLP | RNN | CNN |
|---|---|---|---|
| Data | Tabular data | Sequence data (Time Series,Text, Audio) | Image data |
| Recurrent connections | No | Yes | No |
| Parameter sharing | No | Yes | Yes |
| Spatial relationship | No | No | Yes |
| Vanishing & Exploding Gradient | Yes | Yes | Yes |